

Objective:

To analyze historical sales data from a global electronics retailer using MySQL and Power BI, identify key sales trends, product and category performance, customer and geographic patterns, and generate business insights to support decision-making.

1. What products/categories does the company sell?

The company sells **2,517 products across 8 major categories**. The largest product assortment is in **Home Appliances**, with **661 products**, followed by **Computers** with **606 products** and **Cameras and camcorders** with **372 products**.

The remaining categories are **Cell phones**, **TV and Video**, **Games and Toys**, **Audio**, and **Music**, **Movies** and **Audio Books**.

Overall, **Home Appliances and Computers together account for 1,267 of the 2,517 products, or approximately 50.3% of the company's product catalog**, indicating that these two categories make up a substantial portion of the company's product assortment.

Row Labels	Count of ProductKey
Audio	115
Bluetooth Headphones	50
MP4&MP3	45
Recording Pen	20
Cameras and camcorders	372
Camcorders	103
Cameras & Camcorders Accessories	69
Digital Cameras	100
Digital SLR Cameras	100
Cell phones	285
Cell phones Accessories	30
Home & Office Phones	92
Smart phones & PDAs	101
Touch Screen Phones	62
Computers	606
Computers Accessories	201
Desktops	45
Laptops	78
Monitors	78
Printers, Scanners & Fax	101
Projectors & Screens	103
Games and Toys	166
Boxed Games	46
Download Games	120
Home Appliances	661
Air Conditioners	62
Coffee Machines	74
Fans	78
Lamps	158
Microwaves	102
Refrigerators	86
Washers & Dryers	70
Water Heaters	31

Music, Movies and Audio Books	90
Movie DVD	90
TV and Video	222
Car Video	46
Home Theater System	100
Televisions	50
VCD & DVD	26
Grand Total	2517

2. Which products/categories generate the most revenue and profit?

The analysis shows that **Computers is the company's highest-performing category**, generating approximately **\$19.30 million in revenue and \$11.28 million in profit**. **Home Appliances** is the second-highest-performing category, generating approximately **\$10.80 million in revenue and \$6.30 million in profit**.

At the product level, **desktop computers dominate the highest-revenue and highest-profit products**. Several **WWI** and **Adventure Works** desktop models appear among the top-performing products, while several **52-inch HDTV models** also rank highly.

Overall, the company generated approximately **\$55.76 million in revenue and \$32.66 million in profit**, with **Computers and Home Appliances together contributing about 54% of both total revenue and total profit**.

3. How have sales and order volume changed over time?

The company's sales and order volume showed strong growth from 2016 through 2019, with revenue increasing from \$4.79 million to \$14.66 million and orders rising from 2,865 to 9,083. The strongest year-over-year growth occurred in 2018, when orders increased by 81.86% and revenue by 74.56%. 2019 was the strongest complete year in the dataset. However, performance declined substantially in 2020, with orders falling 48.97% and revenue falling 49.65% compared with 2019. The 2021 figures should not be compared directly with previous full years because the dataset only contains data through February 20, 2021.

4. Are there seasonal patterns?

Yes. The analysis identifies **recurring high-demand periods** around ISO Weeks **1, 7–9, and 51–52** during the complete 2016–2020 period. These periods consistently generated higher revenue and order volume than other weeks.

The effect was present in every year analyzed. Average weekly revenue during the identified high-demand periods was approximately **81% to 217% higher** than during other weeks, while average weekly orders were approximately **85% to 220% higher**. This indicates that the revenue increases were primarily associated with **higher order volume** rather than a major increase in average order value.

The category analysis also shows that the increase is **broad-based**: all eight product categories experienced higher average weekly revenue during high-demand periods. The category with the largest proportional uplift varied by year, so the pattern cannot be attributed to a single product category. However, **Computers remains particularly important because it has the largest overall revenue base and consistently experiences substantial uplift during high-demand periods**.

Business implication

The company can use these recurring periods as historical demand signals for **inventory, replenishment, staffing, and operational planning**. However, the dataset does not contain inventory levels, stockouts, lead times, promotions, or external-event information, so it cannot establish the exact cause of the seasonal pattern or determine precise inventory requirements.

5. Where are customers located?

The company has customers across **five countries: the United States, Germany, Australia, France, and Italy**. The United States has the largest customer base and is also the company's largest market, generating approximately **\$23.28 million in revenue and \$11.14 million in profit**. It also has the highest revenue per customer, at approximately **\$4,080**, indicating strong customer value in addition to its large market size.

At the state/region level, revenue is concentrated in several important markets. **California (\$2.47M), Texas (\$1.76M), New York (\$1.37M), Freistaat Bayern in Germany (\$1.13M), and Florida (\$1.12M)** are among the highest-revenue geographic markets.

The analysis also shows that the company has recorded customers across **all U.S. states, all German states, all Australian states and territories, and the French geographic units represented in the dataset**. Italy has the largest geographic gap, with **13 provinces not appearing in the customer/sales data**: Barletta-Andria-Trani, Caltanissetta, Enna, Fermo, Frosinone, Lodi, Medio Campidano, Monza e della Brianza, Olbia-Tempio, Ogliastra, Carbonia-Iglesias, Verbano-Cusio-Ossola, and Pesaro e Urbino.

Business implication:

The company can use this geographic analysis to focus resources on high-performing markets while identifying regions with limited or no recorded customer presence as potential areas for further expansion research. However, the absence of customers in a region does not by itself prove that there is untapped demand; factors such as market size, customer demand, competition, and distribution feasibility should be evaluated before entering these markets.

6. How long does delivery take, and how has it changed?

Not analyzed due to insufficient delivery-date data. Of the 62,884 sales records, 49,719 (approximately 79.1%) have missing delivery dates, leaving only 13,165 records for reliable delivery-time analysis. Therefore, the available data is insufficient to confidently assess overall delivery performance or changes over time.

7. Which locations, products, or categories are driving performance differences?

The differences in performance across countries are driven by a combination of market size, category mix, and category profitability. Computers are the strongest revenue-generating category in every country, making them a consistent contributor to performance. The United States has the largest and broadest market, with Computers, Cell phones, Home Appliances, and Cameras & Camcorders all contributing substantial revenue.

Category profitability also varies across locations. For example, TV & Video generates revenue in all five countries but records negative profit margins in Australia (-2.54%), Germany (-2.78%), and Italy (-3.12%), while remaining profitable in France and the United States. Other categories such as Audio, Cell phones, and Music, Movies & Audio Books maintain relatively high profit margins across markets.

Overall, geographic performance is influenced not only by how much each market sells, but also by which categories are sold and how profitable those categories are.

8. Which products have high sales but relatively low profitability?

To identify products with high sales but relatively low profitability, products in the top 10% of revenue were classified as high-revenue products, while products with a profit margin below the median product margin of 54.01% were classified as relatively low-profitability. This identified 44 products across five categories.

Computers had the largest concentration, with 25 qualifying products generating approximately \$3.50 million in revenue and \$1.72 million in profit at a 49.02% margin. Home Appliances had 9 qualifying products, followed by Audio with 7. Overall, these products generated approximately \$4.64 million in revenue and \$2.27 million in profit, with an overall margin of about 49.02%.

The key finding is that Computers contain the largest group of high-revenue products whose margins are below the typical product level. These products are still profitable, but their lower relative margins could be an area for the company to investigate through pricing, cost, or product-mix analysis.

9. Are there meaningful differences in order value across sales channels, if the data supports this?

Yes, there is a **small difference in average order value (AOV)** between the two sales channels.

- **In-Store AOV:** \$1,651.94
- **Online AOV:** \$1,602.35
- **Difference:** \$49.59

The in-store AOV is approximately **3.1% higher** than the online AOV. Therefore, the data shows a difference, but it is **relatively small**, suggesting that customers spend fairly similar amounts per order through both channels.

Business insight: The company does not appear to have a major difference in customer order value between online and in-store sales, although in-store orders have a slightly higher average value.

10. If the company launches a new product, which category timing, and market would the historical data support considering?

Based on the historical analysis:

- **Category: Computers** — highest revenue at approximately **\$16.1M** and highest profit at approximately **\$8.05M**.
- **Timing: Saturday** — highest revenue and orders among the days analyzed, with about **\$10.18M revenue** and **6,264 orders**.
- **Market: United States** — highest revenue at approximately **\$18.36M**, substantially higher than the other countries.

Answer:

Historical data suggests that a new product could be considered in the **Computers category**, with marketing and launch activity concentrated around **Saturday**, targeting the **United States market**. This is based on the categories, days, and markets that generated the highest historical sales and revenue.